

E: Determination of the frontal area

Fig. E1 explains one method for determination of the frontal area of a car model. It can of course also be used for a real full scale car.

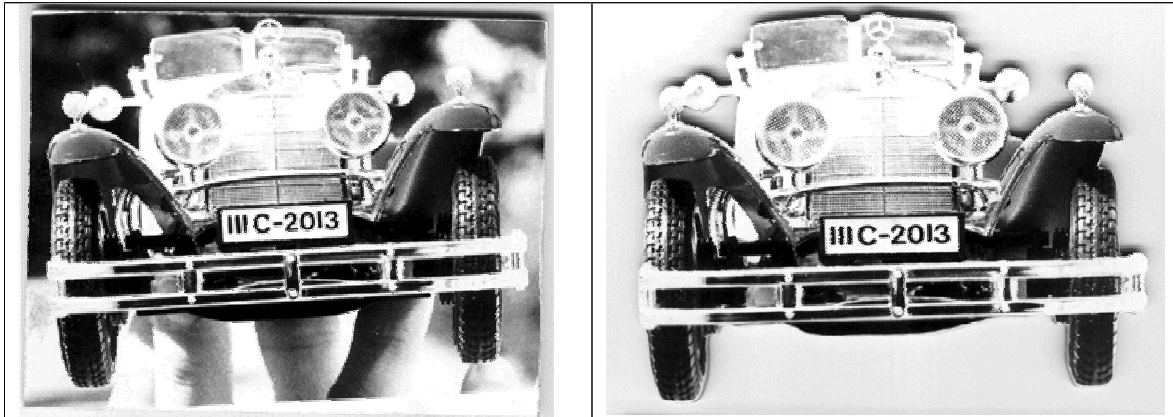


Figure E1: Determination of the frontal area for the model of Mercedes SSK 1928.

Two identical photos are glued onto a piece of plywood. Weighing the first, we can find the mass pr. cm^2 , say σ . From the next, we cut out the area in question by a jigsaw and determine its mass μ , say. The frontal area *on the photo* is then given by μ/σ and for the model it is given by

$$A = \left(\frac{\ell}{\ell_{photo}}\right)^2 \frac{\mu}{\sigma}$$

where ℓ is the length of the front bumper measured on the model, and ℓ_{photo} is the same length measured on the photo.

The photo above is chosen to illustrate some elementary errors. The first of these is easily avoided: It is not taken exactly from the front. The other is slightly more difficult: It is taken at such a close range, that the perspective distort that measures. The windshield, for instance, is too small compared to the bumper. To avoid that sort of error, the photo must be taken from a distance of 2-3 meters, using a telescopic lens.

Having access to a good laboratory weight, it is not necessary to glue the photos onto plywood. One can just use the two photos as they are and cut out the second by a pair of scissors.

A still more simple alternative: Take the photo by a digital camera, open it in a picture manipulating program, select the area by the mouse, and let the program tell the enclosed number of pixels. Compare with the total number of pixels in the photo.